

12 Which statement about two satellites of different masses in the geostationary orbit around the Earth is ***incorrect***?

A The gravitational forces acting on the two satellites are different.

B The centripetal acceleration of each satellite is different.

C The total energy of each satellite is different.

D The angular velocities of the two satellites are the same.

Answer: B

By equating Gravitational force with centripetal force, show that:

$$T^2 \propto R^3$$

where T is the period and R is the radius of the orbital motion.

All geostationary satellites have equal period T, hence all are at the same distance R from Earth.

Since $a_c = R\omega^2 = R\left(\frac{2\pi}{T}\right)^2$, all have equal magnitude of centripetal acceleration.

[OR, Since orbital motion is free fall motion, $a_c = g_{\text{due to Earth at R}} = G\frac{M}{R^2}$, all have equal magnitude of centripetal acceleration.]

Hence Option B is false.

Option A is true because $F_G = F_c = m a_c$. a_c same, but with different m , F_G differs.

Option C is true because it can be shown that Total Energy = $-\frac{1}{2} GMm/R$, which depends on the satellite's mass.

Option D is true because geostationary implies same period, hence same angular velocity.